KAIZEN MATHS

Lesson starter

Further Vectors: Lines, Planes and Products

3D Vectors

Name _____

SA1

Find each requested displacement or position vector.

A has position vector $\begin{pmatrix} 1 \\ -4 \\ -6 \end{pmatrix}$ and B has position vector $\begin{pmatrix} -5 \\ -8 \\ -3 \end{pmatrix}$. Find AB.

SA2

Find the magnitude of each 3D vector.

a = $\begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}$. Find $|a|$.

Show your working

KAIZEN MATHS

Lesson starter

Further Vectors: Lines, Planes and Products

3D Vectors

Name _____

SB1

Find a unit vector in the direction of each vector.

Find a unit vector in the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

SB2

Find the point that divides each line segment in the stated ratio.

P divides AB internally in the ratio 2:2. A = (2, 3, 2) and B = (-1, -12, -1). Find P.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

3D Vectors

Name _____

EA1

Find each requested displacement or position vector.

A has position vector $\begin{pmatrix} -1 \\ -6 \\ -3 \end{pmatrix}$ and B has position vector $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$. Find AB.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

3D Vectors

Name _____

EB1

Find the magnitude of each 3D vector.

$a = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}$. Find $|a|$.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

3D Vectors

Name _____

EC1

Find a unit vector in the direction of each vector.

Find a unit vector in the direction of $\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

3D Vectors

Name _____

EA1

Find each requested displacement or position vector.

A has position vector $\begin{pmatrix} -1 \\ -6 \\ -3 \end{pmatrix}$ and B has position vector $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$. Find AB.

Show your working

Teacher answers & working

Further Vectors: Lines, Planes and Products · 3D Vectors

ALG-ADB4-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SA1

Find each requested displacement or position vector.

A has position vector $\begin{pmatrix} -1 \\ -4 \\ -6 \end{pmatrix}$ and B has position vector $\begin{pmatrix} -5 \\ -8 \\ -3 \end{pmatrix}$. Find AB.

Answer:

$\overrightarrow{AB} = \begin{pmatrix} -4 \\ -4 \\ 3 \end{pmatrix}$

Working:

Use AB = position vector of B − position vector of A.

$\overrightarrow{AB} = \begin{pmatrix} -5 \\ -8 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ -4 \\ -6 \end{pmatrix}$

$\overrightarrow{AB} = \begin{pmatrix} -5 - -1 \\ -8 - -4 \\ -3 - -6 \end{pmatrix}$

$\overrightarrow{AB} = \begin{pmatrix} -4 \\ -4 \\ 3 \end{pmatrix}$

SA2

Find the magnitude of each 3D vector.

$\mathbf{a} = \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}$. Find $|\mathbf{a}|$.

Answer:

$|\mathbf{a}| = 7$

Working:

Use $|\mathbf{a}| = \sqrt{(x^2 + y^2 + z^2)}$.

$|\mathbf{a}| = \sqrt{(-6^2 + 3^2 + 2^2)}$

$|\mathbf{a}| = \sqrt{(36 + 9 + 4)}$

$|\mathbf{a}| = \sqrt{49}$

$|\mathbf{a}| = 7$

Teacher answers & working

Further Vectors: Lines, Planes and Products · 3D Vectors

ALG-ADB4-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SB1

Find a unit vector in the direction of each vector.

Find a unit vector in the direction of $\begin{pmatrix}1\\2\\2\end{pmatrix}$.

Answer:

unit vector = $\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$

Working:

Find the magnitude of the vector.

$|a| = \sqrt{(1^2 + 2^2 + 2^2)} = 3$

A unit vector is found by dividing the vector by its magnitude.

unit vector = $\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$

unit vector = $\begin{pmatrix}0.3333333333333333\\0.6666666666666666\\0.6666666666666666\end{pmatrix}$

SB2

Find the point that divides each line segment in the stated ratio.

P divides AB internally in the ratio 2:2. A = (2, 3, 2) and B = (-1, -12, -1). Find P.

Answer:

$P = (\frac{1}{2}, \frac{-9}{2}, \frac{1}{2})$

Working:

For $AP:PB = 2:2$, use $P = \frac{2A+2B}{4}$.

$P = \frac{2232+2-1-12-1}{4}$

$P = \frac{2-182}{4}$

$P = (\frac{1}{2}, \frac{-9}{2}, \frac{1}{2})$

Teacher answers & working

Further Vectors: Lines, Planes and Products · 3D Vectors

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EA1

Find each requested displacement or position vector.

A has position vector $\begin{pmatrix} -1 \\ -6 \\ -3 \end{pmatrix}$ and B has position vector $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$. Find AB.

Answer:

$\mathrm{AB} = \begin{pmatrix} 3 \\ 6 \\ 4 \end{pmatrix}$

Working:

Use AB = position vector of B − position vector of A.

$\mathrm{AB} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} -1 \\ -6 \\ -3 \end{pmatrix}$

$\mathrm{AB} = \begin{pmatrix} 2 - -1 \\ 0 - -6 \\ 1 - -3 \end{pmatrix}$

$\mathrm{AB} = \begin{pmatrix} 3 \\ 6 \\ 4 \end{pmatrix}$

EB1

Find the magnitude of each 3D vector.

$\mathbf{a} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}$. Find $|\mathbf{a}|$.

Answer:

$|\mathbf{a}| = \sqrt{35} \approx 5.916$

Working:

Use $|\mathbf{a}| = \sqrt{(x^2 + y^2 + z^2)}$.

$|\mathbf{a}| = \sqrt{(5^2 + 3^2 + 1^2)}$

$|\mathbf{a}| = \sqrt{(25 + 9 + 1)}$

$|\mathbf{a}| = \sqrt{35}$

$|\mathbf{a}| = \sqrt{35} \approx 5.916$

Teacher answers & working

Further Vectors: Lines, Planes and Products · 3D Vectors

ALG-ADB4-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EC1

Find a unit vector in the direction of each vector.

Find a unit vector in the direction of $\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$.

Answer:

unit vector = $\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$

Working:

Find the magnitude of the vector.

$|a| = \sqrt{(2^2 + 3^2 + 6^2)} = 7$

A unit vector is found by dividing the vector by its magnitude.

unit vector = $\frac{1}{7}\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$

unit vector = $\begin{pmatrix} 0.2857142857142857 \\ 0.42857142857142855 \\ 0.8571428571428571 \end{pmatrix}$